

Baha Eddine Belhaj Mouldi

Data & AI Engineer

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PROFILE

Computer engineering graduate specialized in data engineering, machine learning and AI systems. Built production-grade data pipelines (900K+ records), BI dashboards, predictive models, RAG/LLM applications and multi-agent architectures. Hands-on with Python, SQL, NLP, sentiment analysis, semantic web and cloud AI services. Seeking Data Engineer, ML Engineer, AI Engineer or Data Scientist roles.

CORE SKILLS

Machine Learning : Python, Scikit-learn, Pandas, NumPy, RandomForest, GradientBoosting, IsolationForest, K-Means, SHAP explainability

Data Engineering : SQL, PostgreSQL, MySQL, MongoDB, Data Warehouse (star schema), ETL pipelines, Pandas, Plotly, Streamlit

NLP & GenAI : RAG (Retrieval-Augmented Generation), LangChain, LLM orchestration, sentiment analysis, text classification, Gemini, Azure AI Foundry

Multi-Agent AI : Agent-to-Agent protocols, distributed agent architectures, workflow orchestration, real-time event correlation

Semantic Web : OWL, RDF, SPARQL, ontology design, knowledge graphs, data interoperability

Visualization & BI : Plotly, Streamlit, Kibana, dashboards, interactive analytics, automated reporting

Tools & Cloud : Python, Git, Docker, Azure AI, AWS, Bash, REST APIs

PROFESSIONAL EXPERIENCE

Junior Cybersecurity Specialist — Stagiaire PFE 01/2026 – 05/2026

Beehive Entreprises, Tunisia

- Built ML-driven detection models (RandomForest, GradientBoosting, IsolationForest) for BeeWAF, achieving 98.2/100 score with 0% false positives on 10,041 detection rules.

Cybersecurity Intern — SAP Threat Detection 07/2025 – 08/2025

Streamlink, Tunisia

- Developed Python data pipelines for SAP log collection, ELK-based analytics dashboards and automated risk scoring for 7 critical transaction types.

Network Security Intern 07/2024

Tunisie Telecom, Tunisia

- Analyzed network traffic data, identified 10+ vulnerabilities, built detection dashboards with Elastic/Kibana and Sigma rules.

DATA & AI PROJECTS

Revenue Assurance Dashboard — Tunisie Telecom 2026

Streamlit, PostgreSQL, Data Warehouse, RAG, Gemini

- Star-schema DW analyzing 900K telecom records across 4 domains. RAG chatbot (Gemini), Plotly dashboards, forecasting (3-30 days), gap detection (NE vs TAX).

Plateforme d'analyse financière — SFBT 2026

Python, Data Warehouse, BI, ML

- DW for financial performance: 20,000+ automated ratios (MSI), BI dashboards, predictive models, end-to-end ETL pipeline.

Multi-Agent Customer Feedback Analysis 2026

Python, NLP, Sentiment Analysis, ML

- Real-time NLP pipeline: sentiment analysis, topic extraction, actionable insights from customer feedback via multi-agent orchestration.

RAG Hotel Recommendation System 2025

Python, RAG, LangChain, LLM, Azure AI

- RAG system: semantic search over reviews, context-aware LLM recommendations, multi-agent orchestration for personalized suggestions.

Multi-Agent Vulnerability Detection System 2025

Python, Multi-Agent, A2A, Azure AI

- Distributed agent architecture (A2A protocol) for real-time event collection, correlation and automated alerting.

Healthcare Ontology 2026

Python, OWL, RDF, SPARQL, Semantic Web

- Semantic ontology: OWL/RDF modeling, SPARQL optimization, interoperability framework for heterogeneous health data.

Web Attack Detection via Data Mining 2024

Python, Scikit-learn, SHAP, AWS

- K-Means clustering (silhouette 0.787) on AWS logs: 39 attacks detected (13.8%), SHAP explainability.

EDUCATION

Engineering Degree in Computer Science 09/2023 – 06/2026

ENIT, Tunisia

Preparatory Classes (Physique-Technologie) 09/2021 – 06/2023

IPEI El Manar, Tunisia

CERTIFICATIONS

Fundamentals of Deep Learning — NVIDIA (2025) ; Exploring Adversarial ML — NVIDIA (2025) ; Machine Learning — NVIDIA (2024) ; Networking — Cisco (2025) ; Git & GitHub — 365 Data Science (2024).

LANGUAGES

English (B2) | French (B2) | Arabic (Native) | Chinese (Basic)